

Confusion Matrix for 1000 products

	Predicted label defective	Predicted label not defective
True label defective	0	380
True label not defective	0	9620

$$\text{accuracy} = \frac{9620 + 0}{9620 + 380 + 0 + 0}$$

$$\text{defective recall} = \frac{0}{0 + 380}$$

$$\text{not defective precision} = \frac{9620}{380 + 9620}$$

$$\text{not defective recall} = \frac{9620}{9620 + 0}$$

Quiz:



		PREDICTED LABEL	
		NEGATIVE	POSITIVE
TRUE LABEL	NEGATIVE	55 TRUE NEGATIVE	5 FALSE POSITIVE
	POSITIVE	10 FALSE NEGATIVE	30 TRUE POSITIVE

Calculate following confusion matrix parameters:

1. Accuracy
2. Precision
3. Recall
4. F1-score

Feature Selection techniques in ML

It is the process of choosing only the most useful input features for a machine learning model.

Helps remove irrelevant and redundant features

Makes models simpler and easier to interpret

model training speed increases

Improves model accuracy and reduces overfitting

Dimensionality reduction techniques





Covariance and Correlation

Covariance

It measures the total variation of two random variables from their expected values.

Using covariance, we can only gauge the direction of the relationship

However, it does not indicate the strength of the relationship, nor the dependency between the variables.

Correlation

It measures the strength of the relationship between variables.

It is dimensionless.



Formula for Covariance

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_j - \bar{Y})}{n}$$

Population Covariance

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_j - \bar{Y})}{n - 1}$$

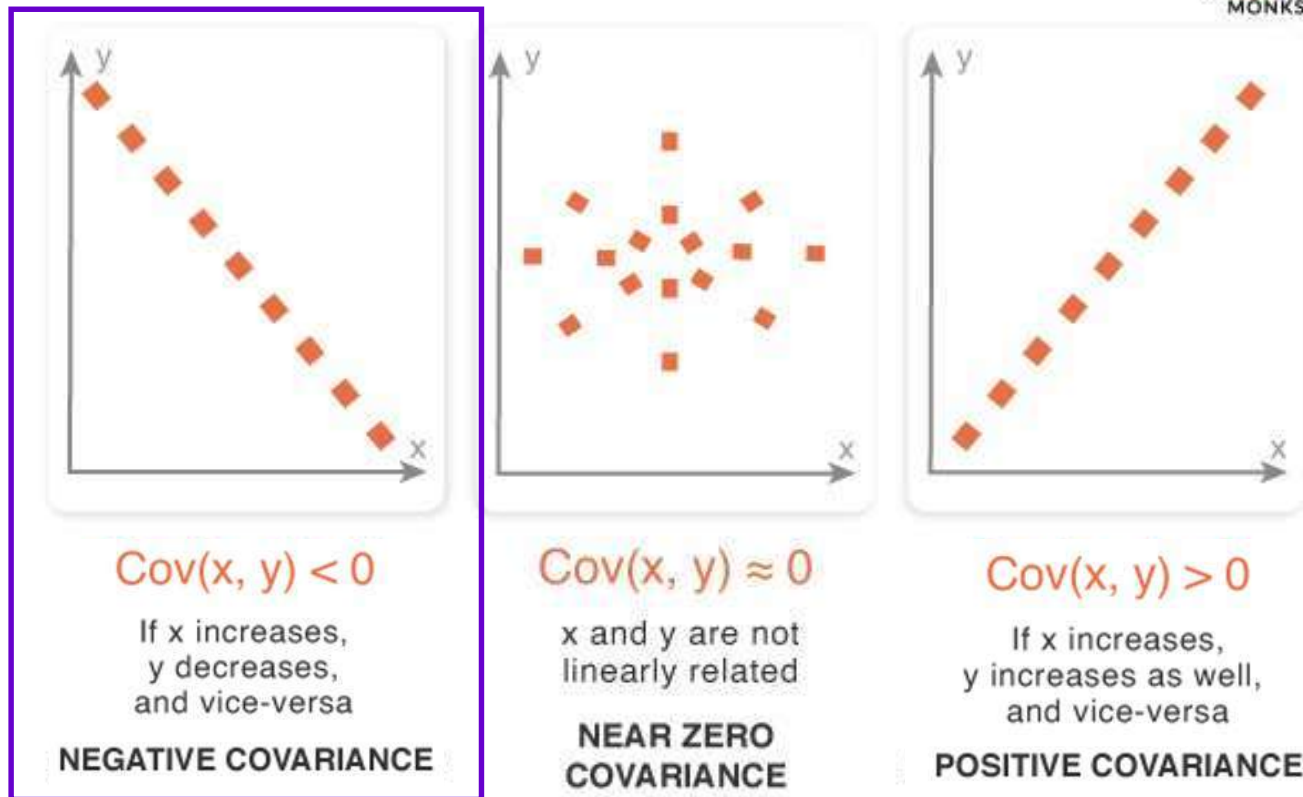
Sample Covariance

Where:

- X_i – the values of the X-variable
- Y_j – the values of the Y-variable
- $\bar{X}$ – the mean (average) of the X-variable
- $\bar{Y}$ – the mean (average) of the Y-variable
- n – the number of data points

Covariance (x,y)

2 variables x (input feature) and y (output feature)





Pearson's correlation coefficient formula (R).

The linear correlation coefficient defines the **degree of relation** between **two variables** and is **denoted by "R"**.

It is also called a **cross-correlation coefficient**

If x & y are the two variables of discussion, then the correlation coefficient can be calculated using the formula

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

n = Number of values or elements

$\sum x$ = Sum of 1st values list

$\sum y$ = Sum of 2nd values list

$\sum xy$ = Sum of the product of 1st and 2nd values

$\sum x^2$ = Sum of squares of 1st values

$\sum y^2$ = Sum of squares of 2nd values



Example 1

Example 2: Calculate the correlation coefficient of the given data.

x	12	15	18	21	27
y	2	4	6	8	12

If x & y are the two variables of discussion, then the correlation coefficient can be calculated using the formula

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Solution:

Here, n = 5

x	12	15	18	21	27
y	2	4	6	8	12
xy	24	60	108	168	324
x ²	144	225	324	441	729
y ²	4	16	36	64	144

$$\sum x = 93$$

$$\sum y = 32$$

$$\sum xy = 684$$

$$\sum x^2 = 1863$$

$$\sum y^2 = 264$$

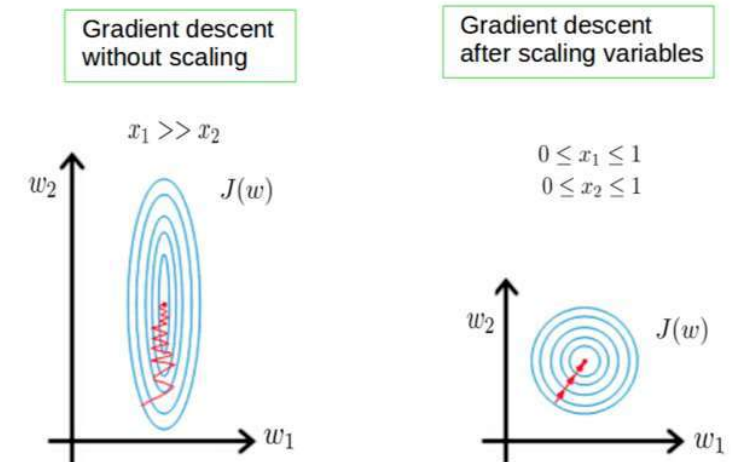
Now, substitute all the values in the below formula.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

$$\text{We have } r = 1.0$$

Feature scaling

- Even after feature selection, the features may not be on the same scale
- This can be problematic when comparing two inputs – features that have larger scales may dominate the result of such comparisons
- Feature Standardization (e.g., by bringing all of them on the same scale such as between 0 to 1)



- Also helpful for stabilizing the optimization techniques used in ML algos

Feature scaling

MinMax Scaler

It scales the data values between 0 and 1

$$x_{\text{scaled}} = (x - x_{\text{min}}) / (x_{\text{max}} - x_{\text{min}})$$

Standard Scaler

It scales the values such that the mean is 0 and the standard deviation is 1

$$x_{\text{scaled}} = (x - \text{mean}) / \text{std_dev}$$

Max Abs Scaler

It takes the absolute maximum value of each column and divides each value in the column by the maximum value.

Robust Scaler | Quantile Transformer Scaler | Log Transformation



Derivatives

Magnitude of derivative at a point is the **rate of change of the func at that point**

sometimes $f'(x)$ to denote the derivative

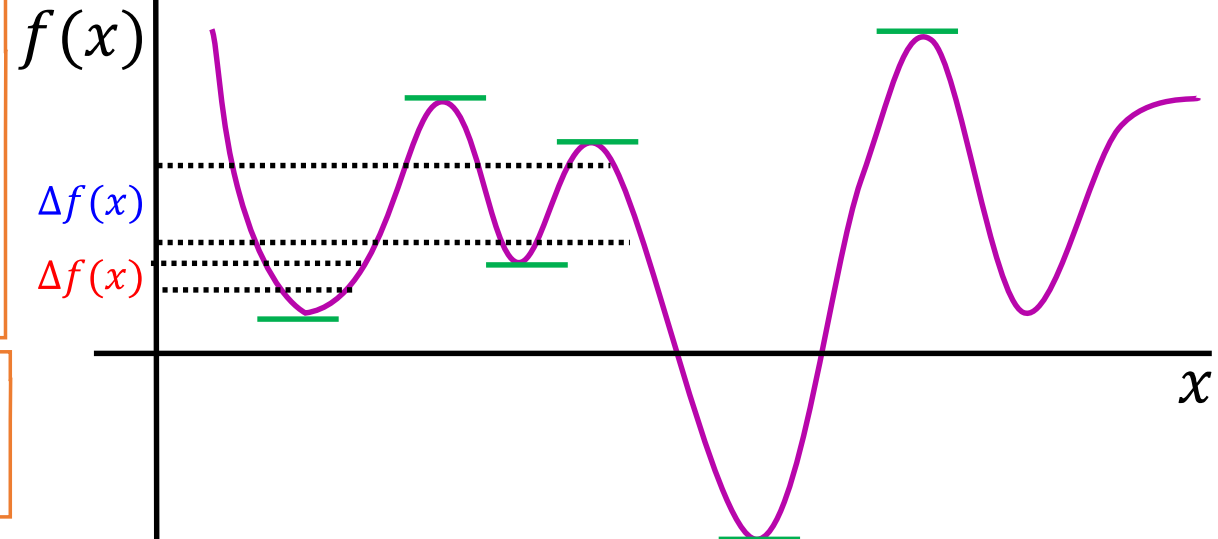
$$\frac{df(x)}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x}$$

Derivative Sign plays an important role in changing the weights and bias of Neural network

Positive derivative means $f(x)$ is increasing at x .

negative derivative means $f(x)$ is decreasing at x

Understanding how $f(x)$ changes its value as we change x is helpful to understand optimization (minimization/maximization) algorithms

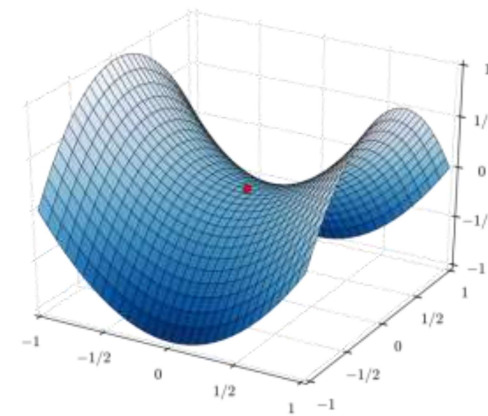
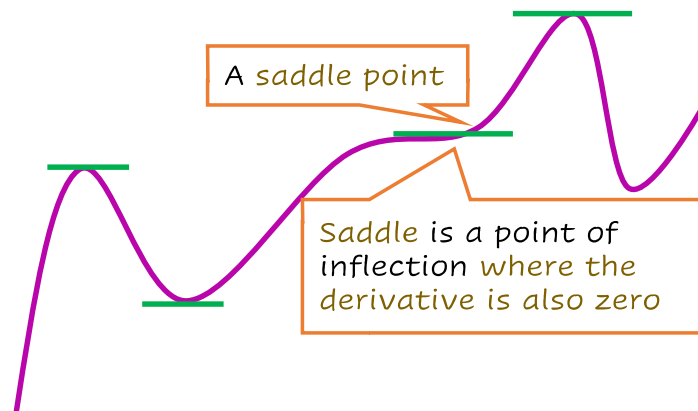


Derivative becomes **zero** at **stationary points** (optima or saddle points)

- The function becomes "flat" ($\Delta f(x) = 0$ if we change x by a very little at such points)
- These are the points where the function has its maxima/minima (unless they are saddles)

Saddle points

Points where derivative is zero but are neither minima nor maxima



Saddle points are very common for loss functions of deep learning models

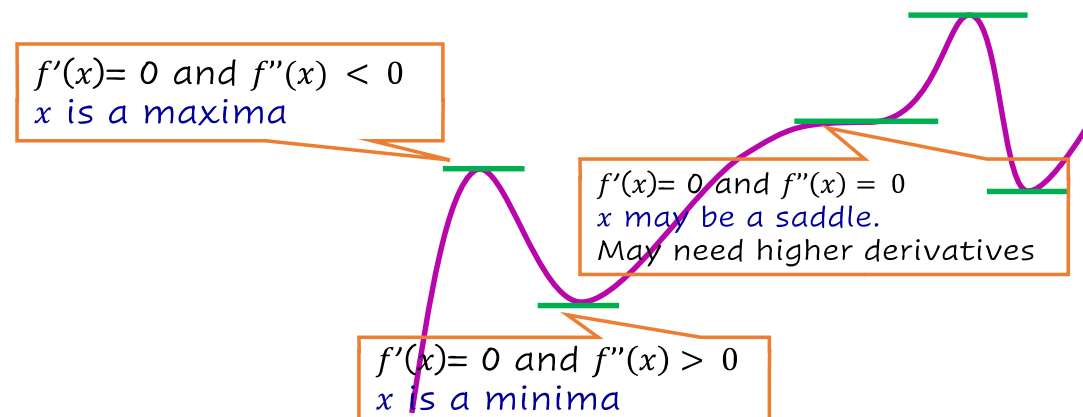
- Need to be handled carefully during optimization

Second or higher derivative may help identify if a stationary point is a saddle

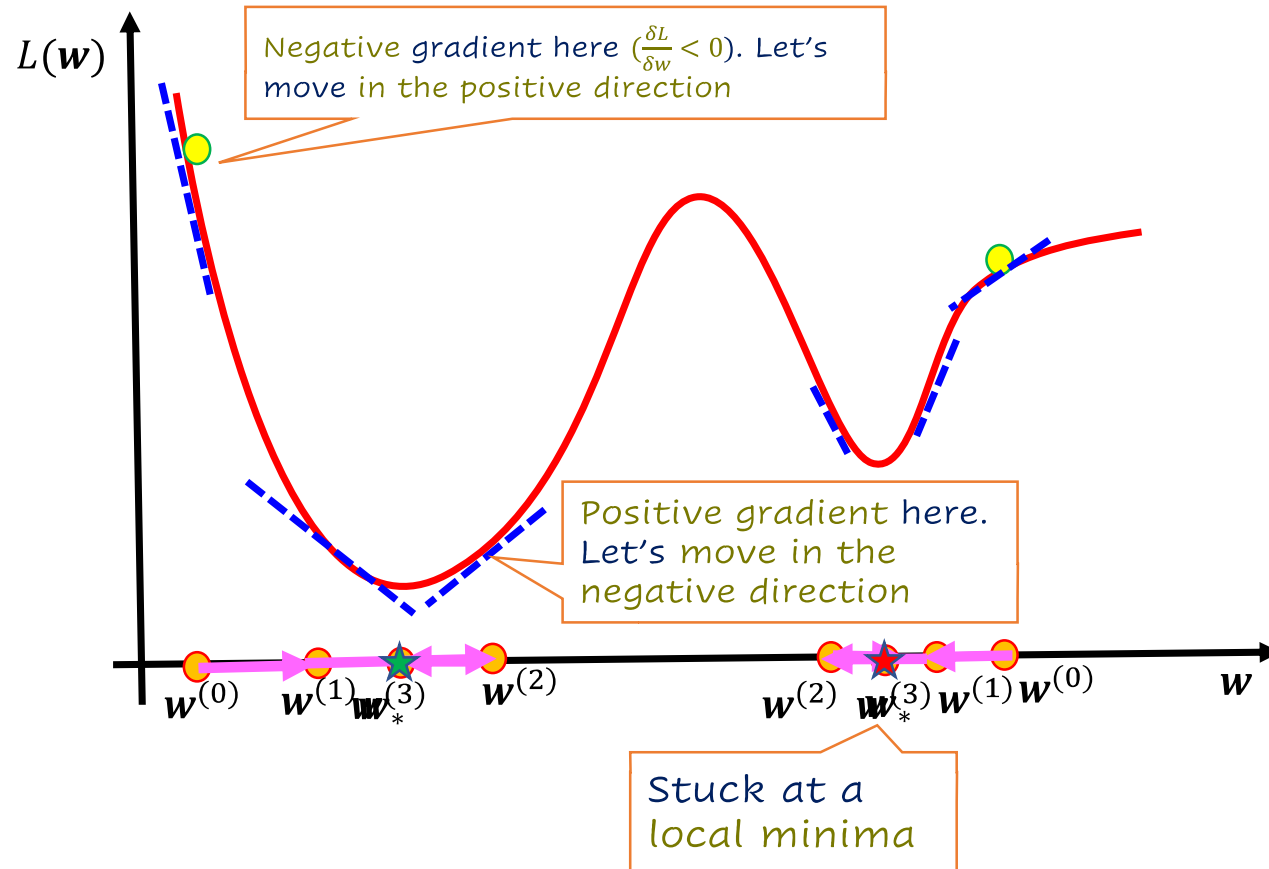
Understanding Optima point through derivatives

How the derivative itself changes tells us about the function's optima?

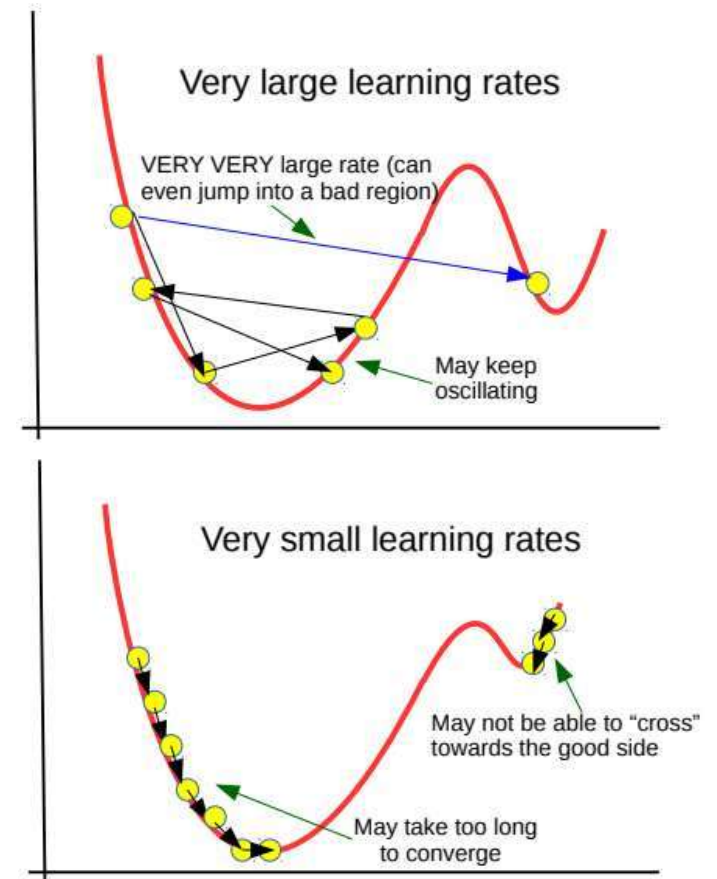
The second derivative $f''(x)$ can provide this information



Gradient Descent Algorithm in Machine Learning



Learning rate is very important

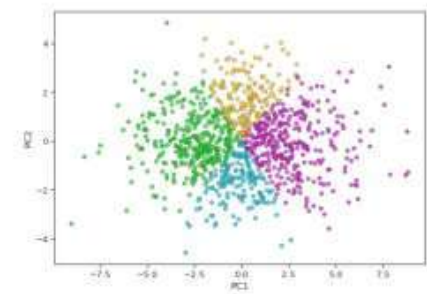
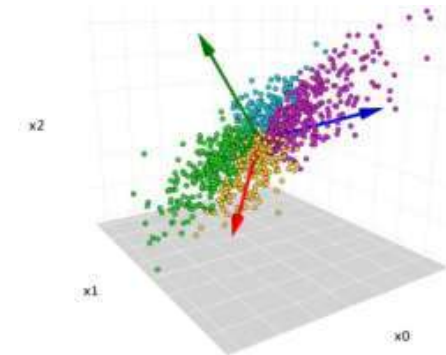


Dimensionality reduction:

Visualizing data with thousands or millions of features, which makes the feature space extremely complex posing problems with regard to training models

Algorithms

Principal Component Analysis (PCA),
nearest neighbors, and
Discriminant analysis.



Dipanjan Sarkar, Raghav Bali, Tushar Sharma

Dimensionality reduction:

Dimensionality reduction:



Feature Selection methods:

Specific features are selected for each data sample from the original list of features and other features are discarded.

No new features are generated in this process.



Feature Extraction methods:

Extract new features from the original list of features.

Thus the reduced subset of features will contain newly generated features that were not part of the original feature set.

PCA falls under this category

PCA was introduced by Karl Pearson, a British mathematician, in 1901 as a statistical method to analyze data variation and relationships

Principle Component Analysis (PCA):

PCA is a powerful technique that helps to simplify large and complex datasets by reducing their dimensionality. It preserves the important information as much as possible.

Salient Features of PCA

PCA is an unsupervised method, meaning it doesn't need labeled data to find patterns.

It helps speed up training and reduce computational costs by working with fewer dimensions.

It creates new features called principal components, which capture the maximum variance in the data.

PCA is useful for data visualization, noise reduction, and feature extraction in machine learning.

it can make features less interpretable, since the new components are combinations of the original ones.

Principle component analysis (PCA)

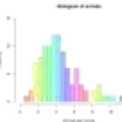
It is a method to find a linear combination that accounts for as much variability as possible

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

Constraint $\alpha_1^2 + \alpha_2^2 + \dots + \alpha_p^2 = 1$

$$\alpha_1^2 + \alpha_2^2 = 1$$

$$0.8^2 + 0.6^2 = 0.64 + 0.36 = 1$$



Diastolic BP (mmHg)	Systolic BP (mmHg)	BP
78	126	138
80	128	140.8
81	127	141
82	130	143.6
84	130	145.2
86	132	148
	Mean	142.7667

$$\text{var}(Y) = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$$= \frac{1}{5} ((138 - 142.8)^2 + \dots + (148 - 142.8)^2)$$

$$= 12.74$$



PCA – Steps with Example

Step 1: Data centering

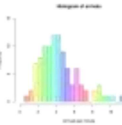
Step 2: Compute the covariance matrix

Step 3: Compute the eigenvalue of covariance matrix

Step 4: Compute the eigenvector of covariance matrix

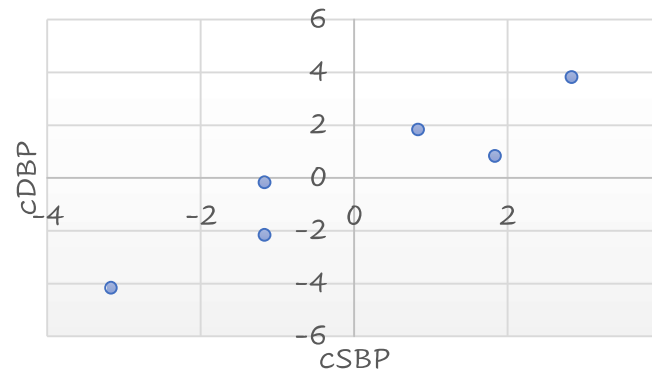
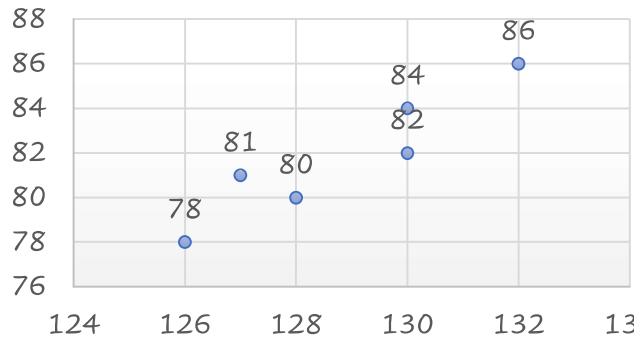
Step 5: Order of eigenvector

Step 6: Compute the principle components



Data Centering

	Systolic BP (mmHg)	Diastolic BP (mmHg)	cSBP	cDBP	cSBP	cDBP
	126	78	2.83	3.83	3	4
	128	80	0.83	1.83	1	2
	127	81	1.83	0.83	2	1
	130	82	-1.17	-0.17	-1	0
	130	84	-1.17	-2.17	-1	-2
	132	86	-3.17	-4.17	-3	-4
Mean	128.83	81.83				



Computation become simpler

Rotation of data is also simpler



Compute Covairance matrix

Sample

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}$$

$$\text{Var}(\text{cSBP}) = 4.4$$

$$\text{Var}(\text{cDBP}) = 8$$

$$\text{Var}(\text{cSBP}, \text{cDBP}) = 5.6$$

	cSBP	cDBP
cSBP	4.4	5.6
cDBP	5.6	8

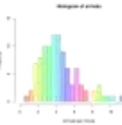
Covariance matrix

$$\det|A - \lambda I| = 0$$

$$\det \begin{bmatrix} (4.4 - \lambda) & 5.6 \\ 5.6 & (8.0 - \lambda) \end{bmatrix} = 0$$

$$3.84 - 12.4\lambda + \lambda^2 = 0$$

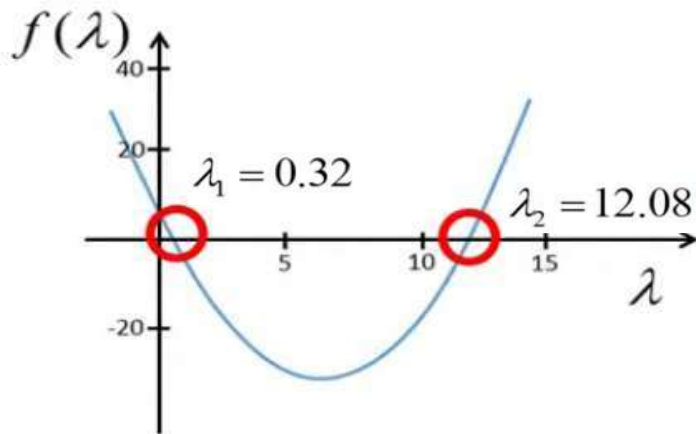
Quadratic Eqn.



Computation of eigenvalues | eigenvectors | Normalization

Eigenvalue

$$3.84 - 12.4\lambda + \lambda^2 = 0$$



Eigenvectors

$$A \cdot v = \lambda \cdot v$$

$$\begin{bmatrix} 4.4 & 5.6 \\ 5.6 & 8.0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 12.08 \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{aligned} 4.4x + 5.6y &= 12.08x \\ 5.6x + 8.0y &= 12.08y \end{aligned}$$

$$5.6y = 7.68x$$

$$5.6x = 4.08y$$

$$y = 1.37x$$

$$v_2 = \begin{bmatrix} 1 \\ 1.37 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix} \quad \lambda_2 = 12.08$$

Normalized Eigenvector



Phyton Code for Eigenvector | Eigenvector

```
[1]: import numpy as np
      from scipy import linalg
```

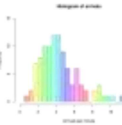
```
[4]: A = np.array([[4.4, 5.6],
                   [5.6, 8.0]])

      # Calculate eigenvalues and eigenvectors
      eigenvalues, eigenvectors = linalg.eig(A)

      print("Eigenvalues:")
      print(eigenvalues)
      print("\nEigenvectors:")
      print(eigenvectors)
```

```
Eigenvalues:
[ 0.31782353+0.j 12.08217647+0.j]
```

```
Eigenvectors:
[[-0.80808699 -0.58906316]
 [ 0.58906316 -0.80808699]]
```



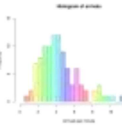
Eigenvector | Eigenvector order

$$\begin{bmatrix} 4.4 & 5.6 \\ 5.6 & 8.0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0.32 \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$v_1 = \begin{bmatrix} -0.81 \\ 0.59 \end{bmatrix} \quad \lambda_1 = 0.32$$

Normalized Eigenvector

cSBP	cDBP
3	4
1	2
2	1
-1	0
-1	-2
-3	-4



Eigenvector order

$$v_1 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix} \quad \lambda_1 = 12.08$$

$$v_2 = \begin{bmatrix} -0.81 \\ 0.59 \end{bmatrix} \quad \lambda_2 = 0.32$$

$$V = \begin{bmatrix} 0.59 & -0.81 \\ 0.81 & 0.59 \end{bmatrix}$$

$$D = \begin{bmatrix} -3 & -4 \\ -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 3 & 4 \end{bmatrix} \quad DV = \begin{bmatrix} -3 & -4 \\ -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0.59 & -0.81 \\ 0.81 & 0.59 \end{bmatrix} = \begin{bmatrix} -5.0 & 0.1 \\ -2.2 & -0.4 \\ -0.6 & 0.8 \\ 0.6 & -0.8 \\ 2.2 & 0.4 \\ 5.0 & -0.1 \end{bmatrix}$$

Principle component score

